Research & Practice for Persons with Severe Disabilities
2011, Vol. 36, No. 1–2, 46–61
copyright 2011 by
TASH

Effects of a Social Skills Intervention Among High School Students With Intellectual Disabilities and Autism and Their General Education Peers

Carolyn Hughes, Melissa Golas, Joseph Cosgriff, Nicolette Brigham,
Caitlin Edwards, and Kelly Cashen
Vanderbilt University

Incorporating general education peers into social skills instructional programs has been effective at increasing social interaction of high school students with intellectual disabilities and autism with their classmates. In this study, communication book use (Hughes et al., 2000), combined with providing the opportunity to interact, was associated with increases in conversational initiations and responses of five high school students identified with intellectual disabilities and autism and their general education peers. We also addressed a limitation of Hughes et al. (2000) by expanding the role of peers during generalization to (a) engage in conversational interactions that were more reciprocal and typical of high school student dyads and (b) provide support to participants while interacting. This study adds to the few published social skills interventions involving high school students with intellectual disabilities and autism in inclusive settings and extends findings to a population of students with more varied communication characteristics.

DESCRIPTORS: autism, high school students, intellectual disabilities, peer support, social skills instruction

Effective social skills are critical to successful school performance, including classroom participation, academic engagement, and social interaction; however, these skills often elude students with intellectual disabilities and autism. Students identified with intellectual disabilities or autism characteristically have limited skills to interact effectively with teachers and peers and display difficulty in responding appropriately to social stimuli (Hendricks & Wehman, 2009; Reichow & Volkmar, 2010). Social skills limitations are exacerbated at the high school level, where conversation with teachers and peers becomes the primary medium for social interaction and class participation and expectations for appropriate communication and requisites to gain social acceptance are raised (Alwell & Cobb, 2009; Wang & Spillane, 2009).

Incorporating general education peers into social skills instructional programs has been effective at increasing social interaction of high school students with intellectual disabilities and autism and their classmates (cf. Carter & Hughes, 2005). For example, Gaylord-Ross, Haring, Breen, and Pitts-Conway (1984) taught three high school students with autism and intellectual disabilities to initiate interaction by introducing a recreational item (e.g., handheld video game) to an array of general education peers during training. Social interactive behaviors generalized to novel peers after instruction was withdrawn and maintained during a 4-month follow-up observation. Davis, Boon, Cihak, and Fore (2010) taught three high school students with autism—one of whom had an intellectual disability—to ask general education peers questions about their individual interests and to listen to peers' responses while maintaining eye contact. Social skills generalized across settings and peers for two of three students with disabilities.

Despite (a) demonstrations of the benefits of peer interaction (e.g., Test et al., 2009), (b) recent legislation (e.g., Individuals With Disabilities Education Improvement Act, 2004; No Child Left Behind Act, 2001) supporting the inclusion of students with disabilities in general education settings and activities to the maximum extent possible, and (c) calls for promoting membership and full participation of students with significant disabilities in general education classes (e.g., Naraian, 2010; Schnorr, 1990), students with intellectual disabilities and autism are among the most segregated of the disability groups. The fact that 44% of students with autism and 52% of students with intellectual disabilities spend 60% or more of their day outside the general education setting (U.S. Department of Education, 2009) indicates the

This research was supported in part by grants from the Organization for Autism Research and the Marino Autism Research Institute.

Address all correspondence and reprint requests to Carolyn Hughes, Department of Special Education, Peabody Box 228, Vanderbilt University, Nashville, TN 37203. E-mail: carolyn.hughes@vanderbilt.edu

limited access to general education activities and peers that is the prevailing model of service delivery and existing practice for these students.

The few observational studies conducted in high school settings indicate that programming to increase communication and social interaction among general education students and their peers with intellectual disabilities and autism rarely occurs and that intervention efforts to increase students' active engagement in general education settings are similarly scarce (e.g., Carter, Hughes, Guth, & Copeland, 2005; Carter, Sisco, Brown, Brickham, & Al-Khabbaz, 2008; Hughes et al., 1999; Mu, Siegel, & Allinder, 2000). Across these studies, researchers found that existing practice shows little evidence of efforts to promote social or classroom participation of students with intellectual disabilities or autism in general education settings and that, consequently, their interaction with general education peers or teachers is extremely limited (e.g., Hughes et al., 1999; Mu et al., 2000). At the same time, few published studies have investigated the effects of interventions to increase social interaction skills of high school students with intellectual disabilities or autism. A recent search of the literature identified only seven studies conducted in high school settings that involved participants with intellectual disabilities and autism and their general education peers. Of these, only three were published in 2000 or later.

Unfortunately, the prevailing practice for supporting students with intellectual disabilities and autism in general education classrooms is via assistance of an adult paraprofessional, which has been shown to further socially isolate these students, particularly at the high school level (Carter & Kennedy, 2006). In addition, pull-out programs, which are commonly used to teach social communication skills to students with significant disabilities, have demonstrated little generalization of target skills (Gresham, Sugai, & Horner, 2001). Involving peers in social skills instruction and support of classroom participation removes the stigma of having an adult paraprofessional assist a student in a general education class. In addition, including multiple peers as supports in students' everyday school settings increases the likelihood that skills will generalize across people, settings, and time. Furthermore, in light of the limited opportunity for interaction and few common experiences that typically exist between students with and without significant disabilities, both social skills instruction and opportunities for social interaction to occur must be provided simultaneously.

This study is a systematic replication of Hughes et al. (2000), in which five high school students with intellectual disabilities, two of whom also had autism, were taught by general education peers to prompt themselves to use communication books to initiate conversation with novel peers across school settings. A total of 4–10 peers instructed each participant as a multiple-exemplar component of the intervention to promote generalization of target skills. Although peers were the sole trainers during the instructional phase, their role was limited during generalization. To allow participants maximum opportunity to initiate conversation, peers were instructed to respond to participants' initiations but were limited to only three initiations during 5-min generalization sessions, limiting reciprocity of interactions. The current investigation addresses this limitation of Hughes et al. (2000) by expanding the role of peers during generalization and promoting conversational interactions more typical of high school student dyads. Specifically, prior to conversational interactions, peers were instructed to expand on topics introduced by participants and to initiate conversation in a more reciprocal fashion than characterized interaction in the previous study. Peers were also instructed to provide support to participants as needed when conversing. In addition, the intervention was introduced as a package in which we taught social skills and provided opportunities for students with intellectual disabilities and autism to use their skills to interact with general education peers.

Research questions were as follows: (a) Could communication book use be adapted to promote more reciprocal and naturalistic conversational interaction between high school students with intellectual disabilities and autism and their general education peers? (b) Could the support role of conversational partners be expanded during generalization? (c) Could the use of communication books be extended to students with a wider range of communication skills (i.e., nonverbal communication to perseverative verbal speech)? (d) Would communication book use generalize across conversational partners and classroom settings when opportunities to interact were provided?

Method

Participants

Participants attended an urban comprehensive high school in a metropolitan school district of 76,000 students in southeastern United States. Total school enrollment was 1196 students (55% Black, 38% White, 4% Hispanic, and 3% Asian and other ethnicities). Forty-five percent of students qualified for free or reduced cost lunches.

Three men and two women were selected from a pool of 25 students enrolled in classes for students with extensive support needs. Selection criteria were (a) a diagnosis of autism or intellectual disability, (b) teacher and parent report of a need to increase social interaction with general education peers, (c) a participant-stated goal to have more friends at school, (d) low or nonexistent of rates of social interaction with general education peers during prebaseline observation, and (e) enrollment in at least two general education classes.

Conner was a 17-year-old White man with Down syndrome, an intellectual disability, and a speech impairment. Conner spoke softly in multiword phrases, although his articulation was poor and often difficult to understand.

At lunch, he interacted with a small group of special education students; however, he rarely was observed to initiate conversation with these peers. Despite attending three general education classes, he infrequently initiated interactions with classmates or teachers.

Joseph was a 20-year-old White man with autism and an intellectual disability. Joseph spoke loudly in full sentences, frequently about weather conditions and almost exclusively to adults. Interactions with general education peers occurred infrequently during prebaseline observations. When Joseph interacted with others, he generally dominated the interactions by engaging in long monologues on topics of his choice or interest, rarely pausing for others to make comments or ask questions. Observers characterized these interactions as "announcements" rather than conversations because Joseph provided information but did not allow others to speak.

Gabby was a 16-year-old White woman who had autism and an intellectual disability. She was largely nonverbal, except for inconsistently responding to adults' questions with a whispered "yes" or one-syllable echolalic utterance. She occasionally followed adults' one-step directions without verbal or physical prompting and responded to their initiations with facial expression or moving toward the speaker. However, Gabby rarely responded to initiations from other students, even when in proximity at lunchtime, and never initiated an interaction during prebaseline observations. Despite her lack of verbal or gestural responses, general education classmates reported that she did occasionally smile. Gabby also displayed behaviors characteristic of autism, such as rocking, avoiding eye gaze, and twirling pieces of string.

Akia was a 21-year-old Asian woman diagnosed with autism, an intellectual disability, and a hearing impairment. During prebaseline observations, she was observed to respond to teachers' one-step verbal or signed directions by signing short answers. Akia was nonverbal and was never observed to vocalize a sound. Most responses were limited to her signing "yes" or "no" to adults that were not coupled with nonverbal social cues, such as smiling or grimacing. She typically displayed a flat affect and showed little apparent interest in peers, never initiating or responding to them.

Lucas was a 19-year-old White man with Fragile X syndrome and an intellectual disability. During prebaseline observations, he rarely spoke, although when he did, he communicated in short, repetitive phrases with poor articulation, often conveying anger or aggression (e.g., "Slam down!"). He rarely interacted with adults or general education students and occasionally made verbal threats or confrontational comments to special education peers (e.g., "I'm gonna hit you!"). He engaged in autistic-like behaviors, such as echolalia, topic perseveration (e.g., wrestling and Georgia Bulldogs football), and listening continually to a favorite CD (e.g., Beach Boys) on headphones. He also displayed social avoidance, such as evading eye gaze and pulling his jacket hood over his head when approached by others. (Additional participant information is displayed in Table 1.)

Conversation Partners

Forty-eight general education students served as conversational partners during intervention, of which 30 were women (19 Black, 8 White, and 3 Hispanic) and 18 were men (11 Black, 6 White, and 1 Hispanic). Number of conversational partners per participant was 16 (Conner), 13 (Joseph), 15 (Gabby), 11 (Akia), and 11 (Lucas); some students served as partners for more than one participant. Selection criteria for conversational partners were an expressed interest in interacting with peers with disabilities and enrollment in the same general education class or lunch period as the participant. For classroom interactions, we asked teachers to identify potential conversational partners, whom we then asked if they would like to participate. For interactions during lunch, we visited the school's community leadership classes (which required community service hours), provided a brief overview of the project, and asked interested students to volunteer to participate during available lunch hours.

Settings

Data were collected in three inclusive school settings for each participant throughout the study, which included the cafeteria during lunch for all participants. Two general education classes in which participants were enrolled were also selected for each participant and included art, physical education, cosmetology, and keyboarding. Baseline and intervention occurred when appropriate within the natural social context of the classroom or cafeteria, such as at the beginning of class, after assigned work was completed, or when students had finished eating and were socializing.

Communication Books

Participants were taught to use communication books to initiate social interactions during the intervention phase (e.g., Hughes et al., 2000; Hunt, Alwell, Goetz, & Sailor, 1990). Communication books were composed of 3×4 in. laminated pages, which were hole-punched and bound. On each page, a Picture Communication Symbol (Mayer-Johnson, 2008) was printed above a typed conversational "opener" identified by Hughes, Harmer, Killian, and Niarhos (1995) and Hughes et al. (1999) as socially appropriate topics derived from a culturally diverse high school population interacting during lunchtime, such as "What class do you have next?" and "What kind of music do you like?"

Communication books contained between 9 and 30 pages, organized to provide a variety of topics (e.g., sports, school events, and recreational interests). Questions were typed opposite each representative symbol to accommodate for differences in participants' communication skills. Because Conner and Joseph could read the questions to their partners, they held their books with

Table 1
Participant Characteristics

Participant	Diagnosis/IQ assessment	Adaptive behavior assessment	Speech/language assessment	Medical/behavioral history	Medication
Conner, 17, White man	Intellectual disability, Down syndrome, speech impairments 42[a]	General adaptive composite parent standard score = 53 General adaptive composite teacher standard score = 61[b]	Speaks in short sentences 52[c]	Speech/language services	None reported
Joseph, 20, White man	Intellectual disability, autism 66[d]	Communication = 63 Daily living = 51 Socialization = 51 Adaptive behavior composite = 54[e]	Not reported	Historical diagnosis of ADHD	None reported
Gabby, 16, White woman	Intellectual disability, autism IQ not available	Communication = 64 Daily living = 73 Socialization = 64 Adaptive behavior composite = 64[e]	57[c]	Speech/language services, non-verbal	Strattera, Klonopin
Akia, 21, Asian woman	Intellectual disability, autism, hearing impairment 66[f]	Communication skills = 40 Daily living skills = 51 Socialization skills = 36 Adaptive behavior composite = 42[e]	Not reported	Speech and hearing impairments, functionally nonverbal, interpreter services, limited social interaction	Abilify, Metformin
Lucas, 19, White man	Intellectual disability, Fragile X syndrome 72[g]	Communication skills = 63 Daily living skills = 64 Socialization skills = 78 Adaptive behavior composite = 66[e]	76[c]	Speech/language therapy, occupational therapy	None reported

Note. [a]Woodcock–Johnson III. [b]Adaptive Behavior Assessment System-II. [c]Peabody Picture Vocabulary Test. [d]Wechsler Adult Intelligence Scale. [e]Vineland Adaptive Behavior Scale. [f]Leiter International Performance Scale. [g]Test of Nonverbal Intelligence (TONI-3).

questions facing themselves during interactions. In contrast, because Gabby, Akia, and Lucas did not read orally, they held their books so that questions faced their partners. Partners would then orally read the question while participants pointed to the accompanying picture.

Outcome Measures

Outcome measures assessed during generalization sessions were (a) percentage of intervals in which interaction occurred between a participant and a partner, (b) percentage of intervals in which a participant initiated or a partner responded, and (c) percentage of intervals in which a partner initiated or a participant responded. *Initiating* was defined as producing a verbal or nonverbal behavior toward a conversational partner that introduced a new topic or expanded on a previous topic (Fey, 1986; Hughes et al., 2000). Initiating included turning to, pointing to, or verbalizing a new conversational opener from a participant's conversational book. *Responding* was defined as producing verbal or nonverbal behavior in response to a conversational partner's initiation (without expanding on a topic or adding new information to a prior utterance) or asking for clarification (Fey, 1986; Hughes et al., 2000). Responding included verbalizations, gestures, or signs. *Conversational topics* were categorized according to topic areas established by Hughes et al. (1999) as topics typically discussed by high school students in informal settings.

Qualitative measures assessed were (a) affect of participants and partners during interaction, (b) quality of interaction, and (c) reciprocity of interaction (e.g., Carter et al., 2005). *Affect* was rated on a 5-point Likert-type scale ranging from 1 (*discouraging of interaction*; e.g., frowning, looking away, and lacking facial affect) to 5 (*encouraging of interaction*; e.g., smiling, maintaining eye contact, and responding frequently). *Quality of interaction* was rated on a 5-point Likert-type scale with poles marked 1 (*low*) and 5 (*high*) based on frequency, duration, and overall affect of interaction between participants and partners. *Reciprocity of interaction* was defined as the degree to which a participant and partner equally initiated conversational interaction. Reciprocity was rated using a 7-point Likert-type scale (1 = *general education partner was primary initiator*, 4 = *general education partner and participant initiated equally*, and 7 = *participant was primary initiator*).

Experimental Design and Conditions

A multiple baseline design across settings and participants (Kazdin, 1982) with a multiple-probe component

(Horner & Baer, 1978) was used to evaluate the effects of communication book use on participants' social interactions. The study consisted of two experimental conditions, (a) baseline and (b) communication book use, during which generalization data were collected daily. Communication book training occurred during the final 2 days (Conner, Akia, and Lucas) or 3 days (Joseph and Gabby) of baseline of each participant's first experimental setting after baseline data were taken for the day. (Asterisks on Figures 1–5 indicate when training sessions occurred.) Follow-up data were collected 4–6 months following the termination of the communication book use condition.

Baseline

During 5-min baseline sessions, we observed a participant and general education peer in closest proximity (i.e., within speaking distance) while they were engaging in expected activities (e.g., completing assignments in class or eating lunch in the cafeteria). No instructional feedback was provided to participants or peers, and the environmental arrangement of the setting was not altered in any way.

Communication book training

During communication book training, the first author trained each participant individually to use a communication book to initiate conversation. Training sessions averaged 10 min (range = 4–22 min). Communication book training began with Conner, with training for the other four participants following sequentially. Using a training script (available on request), the trainer first provided a rationale for training (e.g., "I would like to teach you a way to talk to your friends.") and asked the participant if he or she would like to try. After receiving an affirmative response, the trainer used direct instruction (e.g., modeling, prompting, repeated practice, and

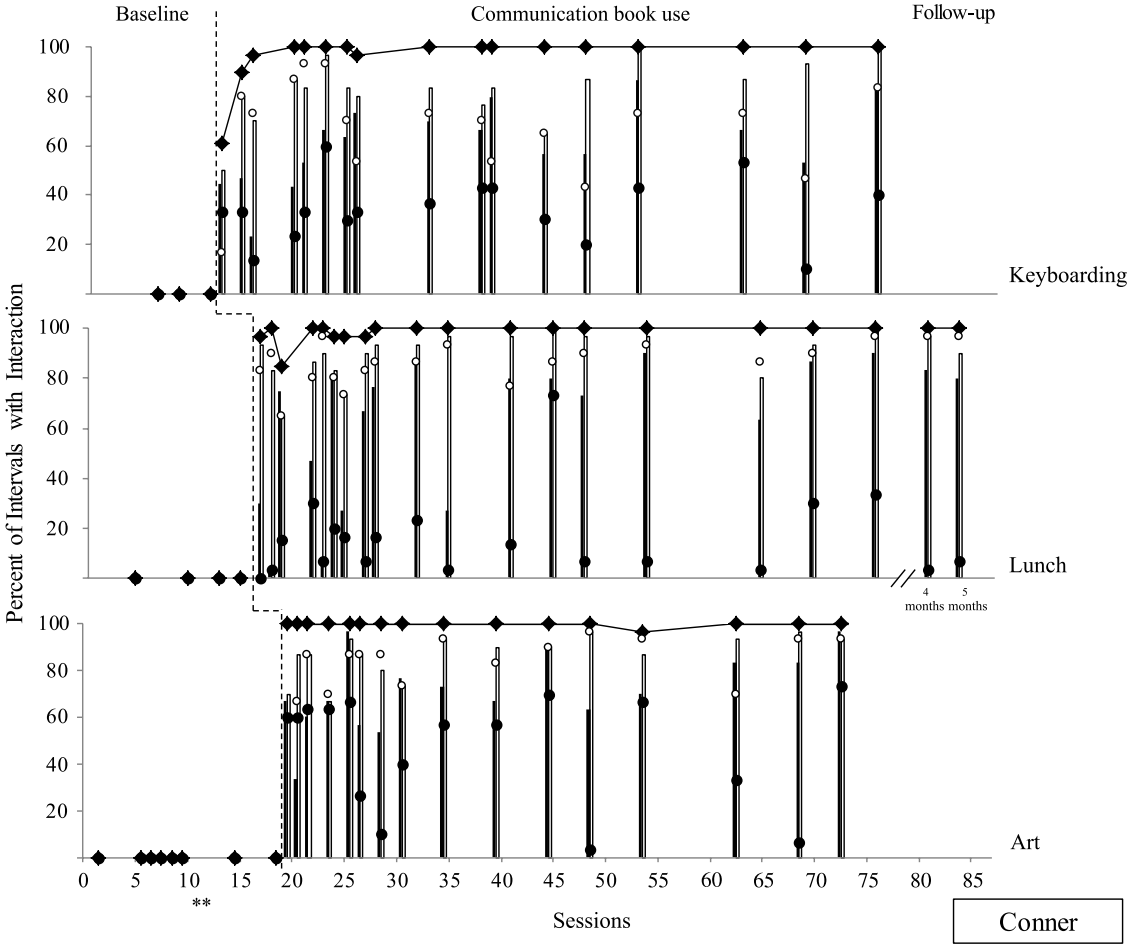


Figure 1. Closed diamonds represent percentage of intervals of overall interaction between Conner and partner during baseline and communication book use. Open circles represent Conner's percentage of intervals initiating, and open bars represent partner's percentage of intervals responding. Closed circles represent partner's percentage of intervals initiating, and closed bars represent Conner's percentage of intervals responding. Asterisks represent training sessions.

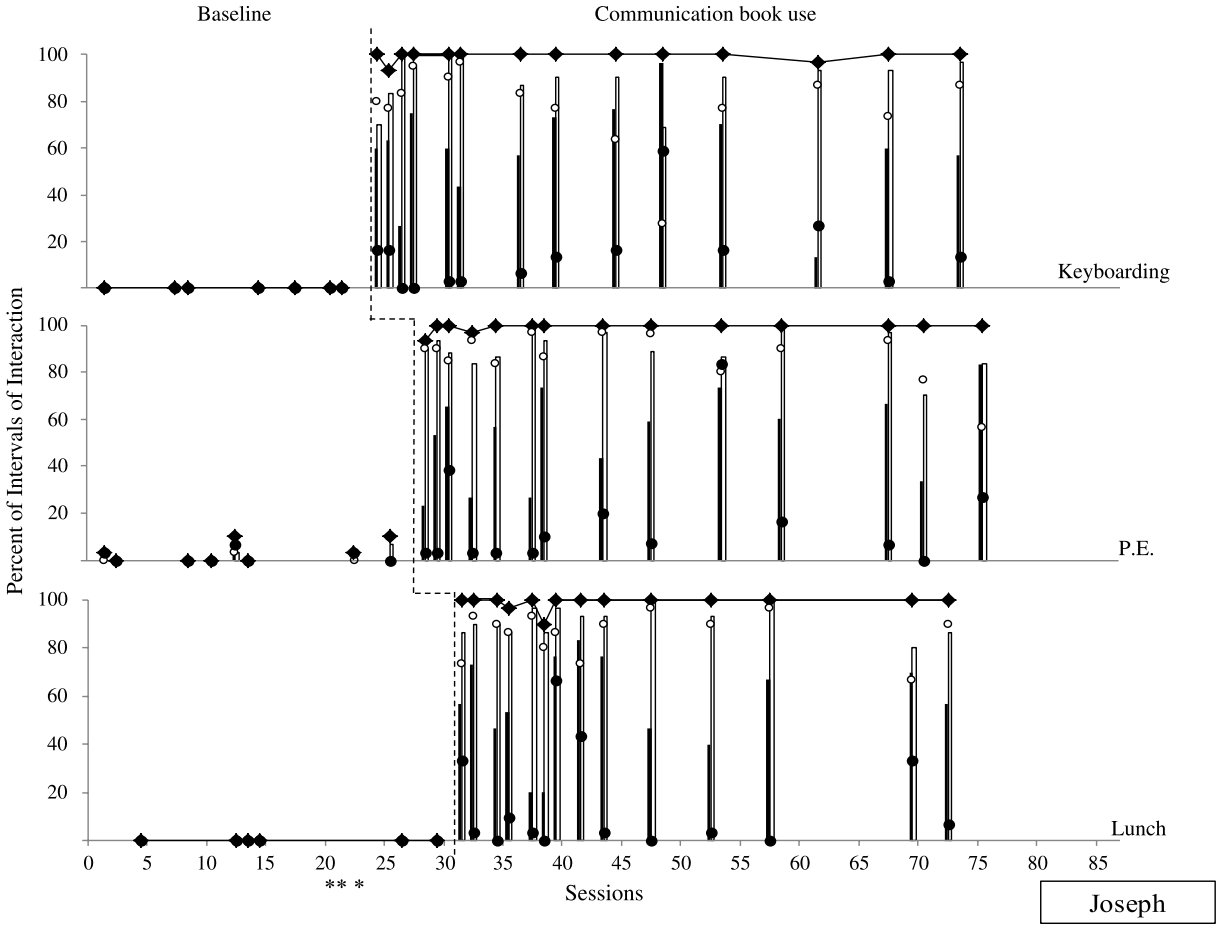


Figure 2. Closed diamonds represent percentage of intervals of overall interaction between Joseph and partner during baseline and communication book use. Open circles represent Joseph's percentage of intervals initiating, and open bars represent partner's percentage of intervals responding. Closed circles represent partner's percentage of intervals initiating, and closed bars represent Joseph's percentage of intervals responding. Asterisks represent training sessions.

corrective feedback) to teach participants to turn the page in a book and read the question (Conner and Joseph) or point to the picture displayed on the page (Gabby, Akia, and Lucas), wait for the trainer to respond, and then turn the page and ask another question. The trainer and participant then practiced conversing by having the participant ask each question in the book (reading the question on the page or pointing to the picture) and the trainer responding to each question asked. The trainer also demonstrated expanding on a topic by introducing new but related information (e.g., saying the names of siblings when asked, "Do you have any brothers or sisters?"). At the end of each session, the trainer told participants to use their books to talk to their friends. (Training data are available on request.)

Communication book use

Training was completely withdrawn for Conner, Joseph, Gabby, Akia, and Lucas on Sessions 12, 24, 35, 47, and 59, respectively, to assess generalization of conversational interaction when an opportunity to converse with a peer was provided. At the beginning of each session, we asked a participant if she or he would like to talk to a general education peer, who had previously been trained as a conversational partner (see below). (Although no participant refused, consent was required for an interaction to occur.) Upon consenting, the participant walked to the partner's location if the two were not already in proximity. The two students were introduced if they did not previously know each other. To minimize researcher assistance and prompt peer engagement, we handed the conversational booklet to the partner who then handed it to the participant while the two exchanged greetings. No prompting or instructional feedback was provided.

Conversational Partner Training

Immediately prior to conversational partners' first interaction with a participant and in the same experimental setting, they were informally trained on communication book use. An observer briefly modeled question-asking,

using the book and possible responses. Observers also demonstrated how partners could expand on topics and ask reciprocal questions. When paired with participants who did not read (Gabby, Akia, and Lucas), partners were instructed to verbalize the question on a card prior to answering. Partners were also taught to prompt turning to a new card if a participant did not independently and provide additional support as needed, such as ad-libbing conversation, smiling, and showing signs of affirmation.

Observation and Recording Procedures

Participants and their partners were observed once daily during 5-min sessions during baseline and communication book use conditions. We used a 10-sec partial interval recording system to assess initiations and responses. Conversational topics were tallied as they occurred. Immediately following a session, (a) student affect and (b) quality and reciprocity of interaction were rated if an interaction had occurred. During training sessions, event recording was used to measure participants' initiations, responses, and communication book use.

Observers and Observer Training

Eight graduate students in special education served as observers. Prior to data collection, all observers read and discussed behavior code definitions, rules for scoring, and observation procedures. Observers then practiced observation and recording procedures by watching videotapes of students conversing, followed by *in situ* coding. Discrepancies in coding were discussed until agreement was reached. Observers were required to reach a criterion of 80% interobserver agreement for all outcome measures for two consecutive practice sessions before collecting data.

Interobserver Agreement

Interobserver agreement was assessed during a minimum of 33% of baseline and 50% of communication book use sessions per participant per condition. The point-by-point agreement method (Kazdin, 1982) was used to assess percentage agreement for all dependent measures. Overall interobserver agreement means and ranges were overall interaction (98%, range = 98–100%),

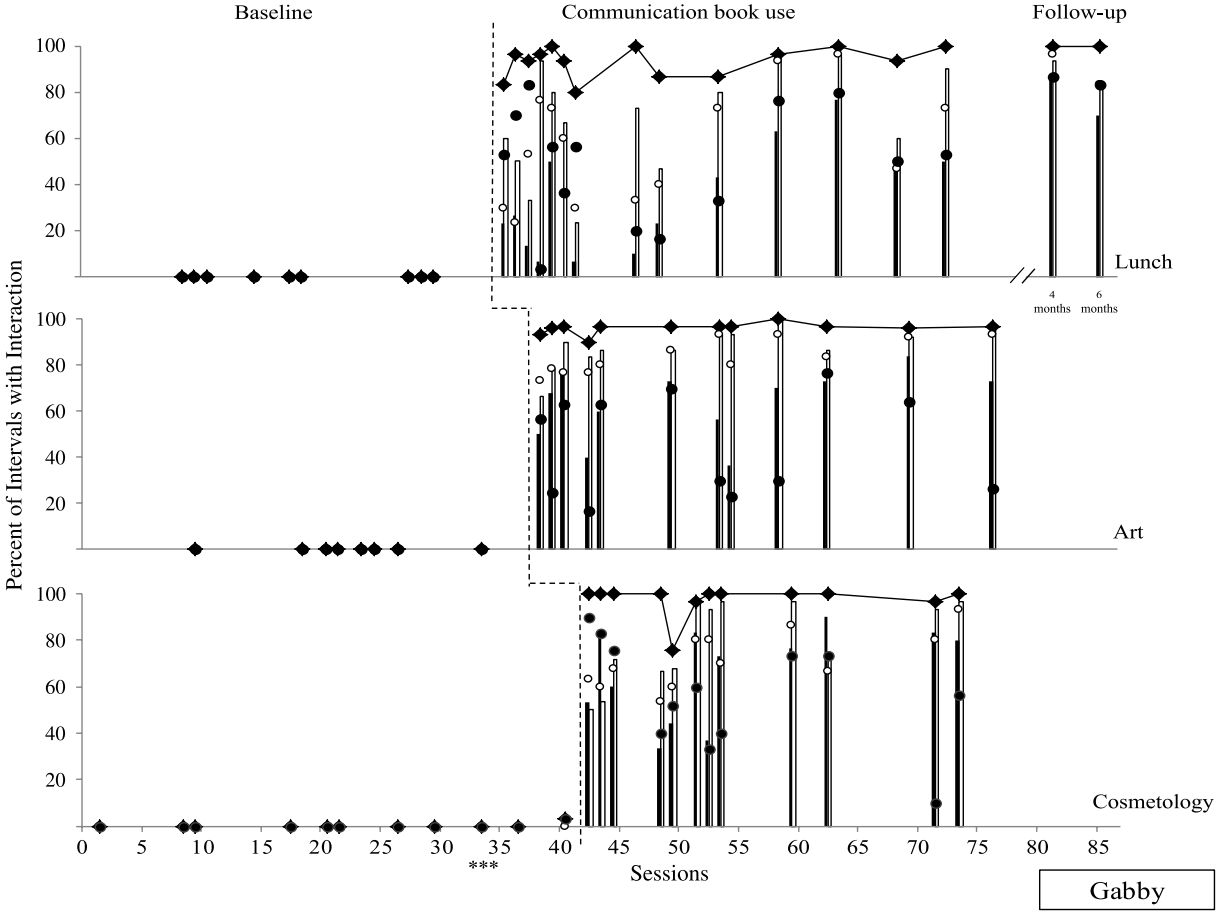


Figure 3. Closed diamonds represent percentage of intervals of overall interaction between Gabby and partner during baseline and communication book use. Open circles represent Gabby's percentage of intervals initiating, and open bars represent partner's percentage of intervals responding. Closed circles represent partner's percentage of intervals initiating, and closed bars represent Gabby's percentage of intervals responding. Asterisks represent training sessions.

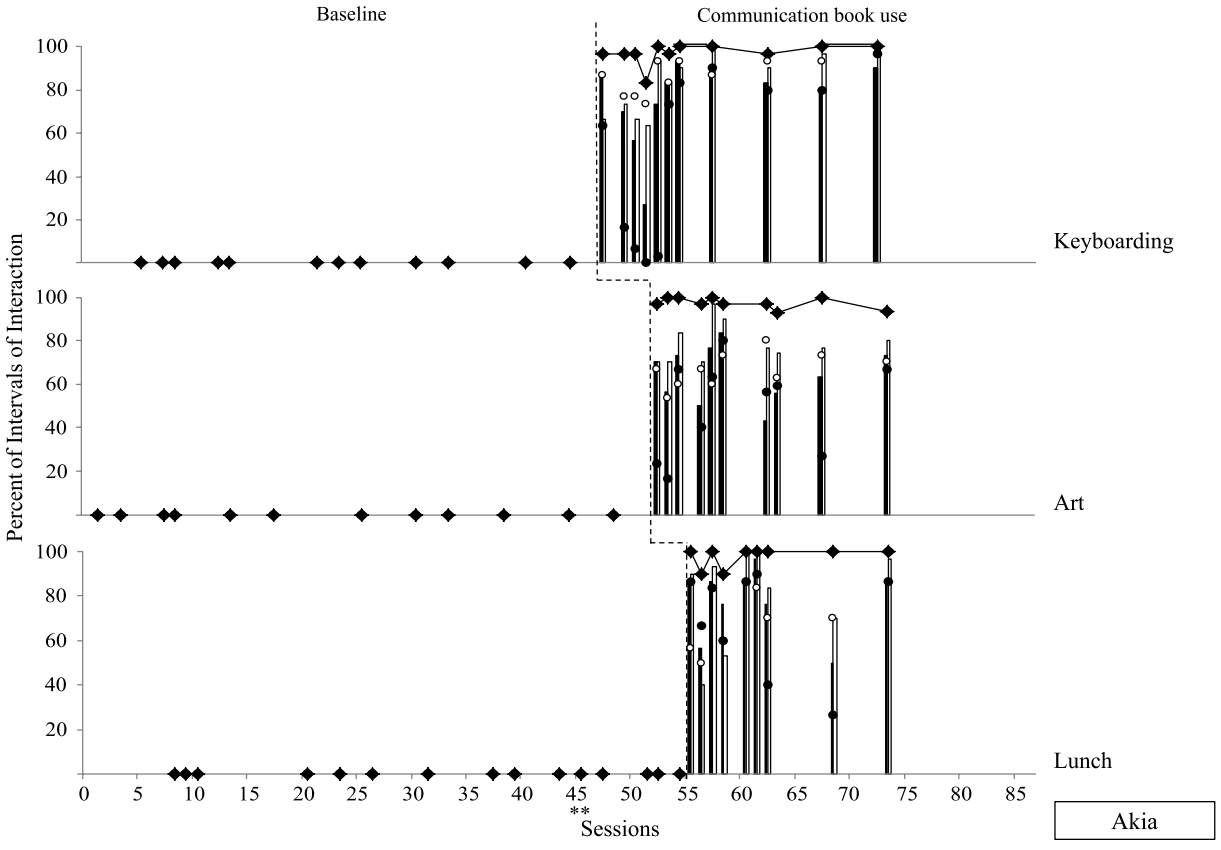


Figure 4. Closed diamonds represent percentage of intervals of overall interaction between Akia and partner during baseline and communication book use. Open circles represent Akia's percentage of intervals initiating, and open bars represent partner's percentage of intervals responding. Closed circles represent partner's percentage of intervals initiating, and closed bars represent Akia's percentage of intervals responding. Asterisks represent training sessions.

participants' initiations (86%, range = 43–100%), partners' responses (89%, range = 47–100%), partners' initiations (85%, range = 43–100%), participants' responses (82%, range = 40–100%), participants' affect (99%, range = 75–100%), partners' affect (99%, range = 50–100%), quality of interaction (99%, range = 75–100%), and reciprocity of interaction (96%, range = 75–100%).

Social Validation Measures

Social validation measures (Wolf, 1978) were collected to assess the importance and acceptability of program goals, procedures, and outcomes. Prior to collecting data, we queried participants about their social goals (e.g., "Would you like to have more friends at school?" and "Would you like to have more friends in _____ class?"). We then asked participants at the conclusion of the study if they believed they achieved their goals and if their books had helped them talk to new friends. Following communication book use sessions, conversational partners were asked to complete written questionnaires in which they evaluated their interactions with participants. At the end of the study, we also asked partners to evaluate the acceptability and effectiveness of the communication book.

Because a normative range of opportunity for and occurrence of social interaction among general education students in high school classes has not been established in the literature, we observed 135 random dyads of students under baseline conditions (no intervention) using the same outcome measures applied to participants. We collected data during 15 5-min observational sessions in each of 9 general education classes attended by participants for a total of 135 observations (11.25 hr). Of the 270 students observed, 172 were Black, 70 were White, 22 were Hispanic, and 6 were other ethnicities; 155 (57%) were women.

Fidelity Measures

We measured fidelity of training during all training sessions across participants using a 10-item training script. Mean training fidelity was 98% (range = 80–100%). We measured fidelity of the intervention during 82% of communication book use sessions using a 12-item checklist of intervention components (e.g., consent of participant is

obtained, booklet is handed to partner). Mean fidelity during intervention was 100% (range = 90–100%).

Results

Generalization Sessions

Percentage of intervals interacting

Figures 1–5 (closed diamonds) show percentage of intervals in which participants and conversational partners interacted across baseline and communication book use sessions across settings. During baseline, participants rarely interacted with general education students (M = 0.23% of intervals of interaction per session across participants, range = 0–1.27%). Introduction of communication book use was followed by immediate increases in percentage of overall interaction for all participants across settings, which maintained over time. Following two training sessions for Conner, Akia, and Lucas and three for Joseph and Gabby, mean percentage of intervals in which interaction occurred across communication book use sessions was 98%, 99%, 96%, 97%, and 99% for Conner, Joseph, Gabby, Akia, and Lucas, respectively. Little variability in overall interaction was found across settings for participants.

Percentage of intervals in which participants initiated or partners responded

Open circles on Figures 1–5 show percentage of intervals in which participants initiated to their conversational partners. During baseline, only Joseph (four 10-sec intervals) and Lucas (one 10-sec interval) initiated conversation to general education peers with a total of only 4 10-sec intervals of partner responses (open bars) across 145 5-min observation sessions across the 5 participants. After participants were trained to use their communication books, all increased their percentage of intervals initiating to within the previously established range of expected behavior for general education high school students conversing informally (30–80% time initiating; Hughes et al., 1995, 1999). Mean initiations were 79%, 84%, 71%, 76%, and 81% for Conner, Joseph, Gabby, Akia, and Lucas, respectively. Mean partner

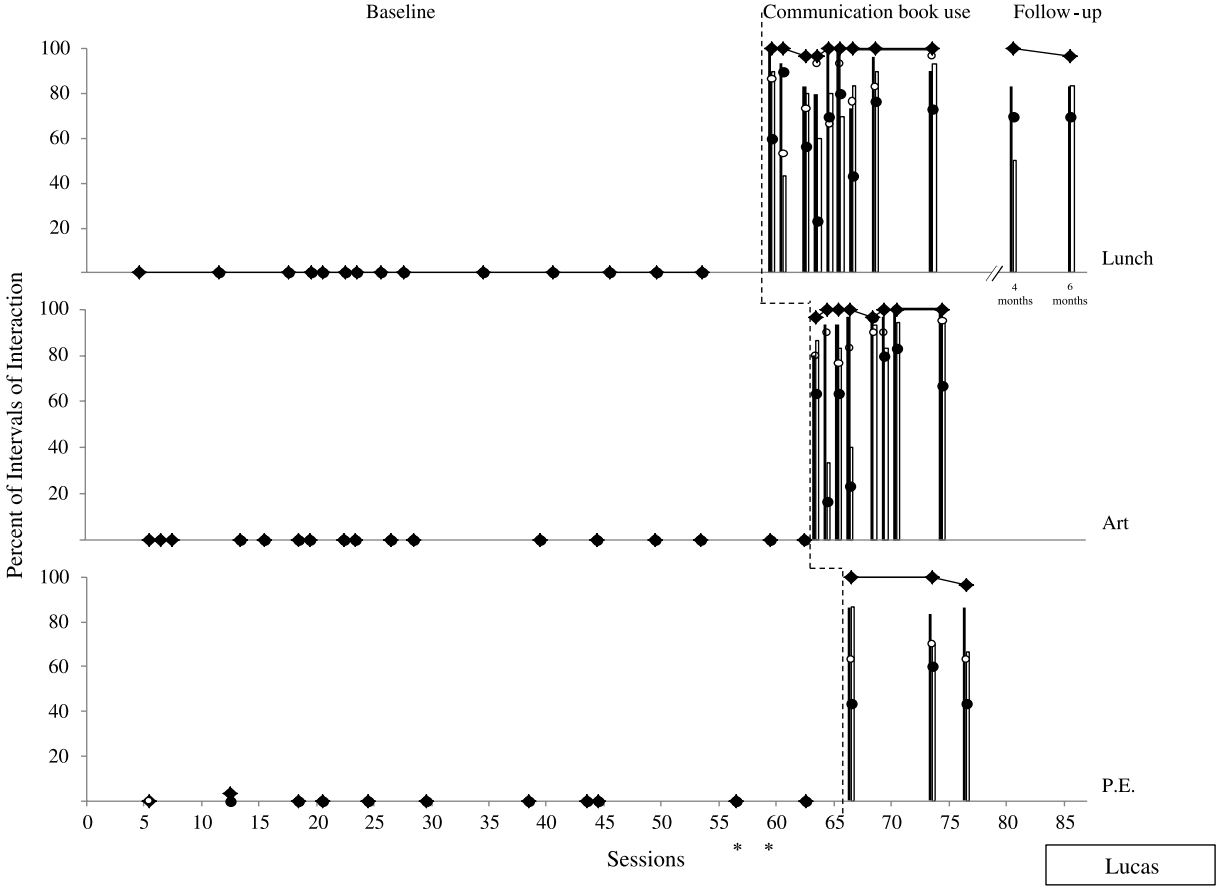


Figure 5. Closed diamonds represent percentage of intervals of overall interaction between Lucas and partner during baseline and communication book use. Open circles represent Lucas's percentage of intervals initiating, and open bars represent partner's percentage of intervals responding. Closed circles represent partner's percentage of intervals initiating, and closed bars represent Lucas's percentage of intervals responding. Asterisks represent training sessions.

responses systematically increased to a range of 75% (Lucas) to 90% (Joseph), suggesting that dyads were engaged in conversational turn taking.

Percentage of intervals in which partners initiated or participants responded

Closed circles on Figures 1–5 show percentage of intervals in which partners initiated to participants. A general education peer initiated to a participant during only five 10-sec intervals with a total of only three 10-sec intervals of participant responses (closed bars) across all baseline sessions. During communication book use, percentage of intervals in which partners initiated increased across all participants, although percentage of initiating varied considerably across participants. Specifically, mean partner initiating to Gabby, Akia, and Lucas was 51%, 57%, and 60%, respectively, versus 32% and 15%, respectively, to Conner and Joseph. Participants' responding also varied from a mean of 53% (Gabby) to 92% (Lucas). Type of participant responses also varied from full sentences (Joseph) to short phrases (Conner and Lucas) to nods or whispered "Yes" (Gabby) and signed "Yes" (Akia), which accounted for differences in percentage of intervals responding across participants (e.g., Joseph typically responded at length to initiations vs. Gabby's abbreviated responding). In general, partners tended to initiate more frequently to participants whose verbal skills were more limited and who pointed to, rather than read, conversational book questions to partners.

Conversational topics discussed

Conversational topics discussed most frequently by participants and partners during communication book use were nonschool events (e.g., holidays or/and afterschool activities), academically related school events (e.g., teachers or classes), and socially related school events (e.g., sports or pep rallies), which all fell within the categories of typical high school conversation identified by Hughes et al. (1999). Mean number of topics discussed per 5-min session was 5 (range = 3–8), indicating that a variety of expected topics among high school students was discussed among participants and partners when conversing.

Affect ratings

Participants generally received positive affect ratings throughout communication book use. Mean affect ratings were 4.7 (Conner), 4.3 (Joseph), 3.9 (Gabby), 3.8 (Akia), and 4.7 (Lucas). The greatest range in ratings was for Gabby (2–5), who also had the lowest mean rating in a setting (3.4 in lunch). Gabby also had the lowest percentage of initiating (58%) and responding (31%) of all participants in that setting. Lucas never had an affect rating below 4 and Akia was the only participant to never receive a rating of above 4. Therefore, although students who were nonspeaking (Gabby and Akia) tended to receive slightly lower affect ratings, pointing to versus reading questions to a partner (Lucas) did not deflate affect ratings. Interestingly, affect ratings of conversational partners were found to be consistently higher with those participants who had limited verbal skills and who pointed to rather than read questions. Specifically, partners' ratings never fell below 4 when they interacted with Gabby, Akia, and Lucas. Mean affect ratings for partners were 4.5 (Conner), 4.1 (Joseph), 4.7 (Gabby), 4.5 (Akia), and 4.9 (Lucas).

Quality and reciprocity of interaction

Mean quality of interaction ratings were generally high across participants, with the highest ratings for Lucas (4.9) and Conner (4.6), who also had the highest affect ratings. Ratings were slightly lower for Gabby and Akia (4.2) and for Joseph (4.1). The lowest and most variable ratings across settings were found in lunch for Gabby (3.7) and Joseph (3.9). In contrast, reciprocity of interaction was found to vary considerably across participants. Mean ratings for Joseph (5.6 of a possible 7.0) and Conner (5.1) indicated that they had a greater role in guiding interactions than did their partners. These students also read rather than pointed to their conversational questions. Lucas's mean rating of 4.6 indicated a fairly equal role between participant and partner in guiding the conversation. Participants with very limited verbal skills (Akia and Gabby) had scores that indicated that partners had the greatest role in guiding interactions (2.5 and 3.5, respectively).

Participants' Social Goals

Participants' responses to pre- and postintervention interview questions addressing their social goals are summarized in Table 2. During their preintervention interviews, all participants indicated verbally or by gesturing or signing that they would like to have more friends at school to sit with at lunch and in different classes. Responses ranged from "happy" to "scared" when meeting someone new and "be nice" when asked what they could do to have more friends. During postintervention interviews, all students indicated verbally or by signing (Akia) that they now had more friends at school in their classes or at lunch. All participants also responded that the communication books helped them talk to their new friends and they enjoyed using the book (e.g., "[The book] helped me talk about other things instead of the same thing over and over because that bores people sometimes.")

Social Validation

Conversational partners completed questionnaires immediately following 38% of intervention sessions across participants, in which they rated the enjoyableness and similarity of their interactions to those with their friends. Questionnaires were composed of five items rated on a 5-point Likert-type scale with poles indicating *least favorable* (1) to *most favorable* (5) response. Mean responses across all items and questionnaires (N = 69) was 4.1 (range = 1–5; see Table 3), indicating that partners generally had positive perceptions of their interactions. The

Table 2
Participants' Social Goals

Participant	Preintervention question	Postintervention question
	Would you like to have more friends at school?	*Do you have more friends at school now?*
Conner	"Yes."	"Um, yes."
Joseph	"Yeah."	"Um, yeah. I feel like I have some more now."
Gabby	"Yes."	"Yes."
Akia	"Yes." *(Signed)*	"Yes." *(Signed)*
Lucas	"That's fine."	"Yeah, hang out."
	Would you like to have friends to sit with at lunch who are from other classes?	*Do you have more friends to sit with at lunch now who are from other classes?*
Conner	"Yes."	"Um, I think C___ and K___."
Joseph	"Yeah, I guess."	"I don't know if that's changed a whole lot. But, I do feel like I have more people to talk to like in the hallway."
Gabby	"Yeah."	"Yes."
Akia	"Yes." *(Signed)*	"Yes." *(Signed)*
Lucas	"No."	"No, my old friends."
	Would you like to have more friends when you go to different classes?	*Do you have more friends now when you go to other classes?*
Conner	"Yes, I want different more friends, helpful."	"Um, J___, M___ in art."
Joseph	"Yeah."	"I feel like I have more people to talk to. It's neat getting to know new people. It feels special when you know a lot of people."
Gabby	"Yes."	"Yes."
Akia	"Yes." *(Signed)*	"Yes. Friends talk." *(Signed)*
Lucas	"No, no."	"Yeah, A___, J___, my buddy."
	How do you feel when you meet someone new?	*How do you feel when you talk to your new friends?*
Conner	"New different people, I think I feel people's heart *(points to heart)*. I am excited."	"It make me feel sort of happy and glad."
Joseph	"I guess, kind of a little different. I guess I like the idea of getting to meet someone new."	"It's different. A little nervous. But, it's neat because I get to know their lives, the things they like and dislike."
Gabby	*(No response)*	*Looked at general education peer and waved.*
Akia	*Pointed to* "Happy." *(Given choice of happy or sad face)*	"Yes." *(Signed)*
Lucas	"I don't know…Scared."	"Good, okay."
	Tell me some things you could do to have more friends at school.	*Does your new book help you talk to your new friends?*
Conner	"She say, he say."	"Yes, I talk and say, 'What's up?'"
Joseph	"Just do the right things, stay out of trouble-that would help."	"I think it definitely helped. I knew how to talk before but it helped me make new friends. It also helped me talk about other things instead of the same thing over and over. Because that bores people sometimes-like the weather."
Gabby	*(No response)*	"Yeah."
Akia	"No cry." *(Signed)*	"Yes." *(Signed)*
Lucas	"Be nice."	"Yeah."
	N/A	*Do you like to talk to your new friends using your book?*
Conner		"Uh-huh. Because I feel glad in my heart."
Joseph		"Yeah, I do think in general. I'd like to start talking without the book."
Gabby		"Yeah."
Akia		"Yes." *(Signed)*
Lucas		"Yeah."

lowest ratings were found for Akia, which may have related to her characteristically flat affect and lack of speech. Pointing to rather than reading questions did not appear to deflate partners' reactions to conversing with Lucas, who received the highest mean rating overall. Peers' open-ended comments on questionnaires also revealed favorable views toward their interactions with participants, such as "Conner is more prone to just walk up to me and start a conversation on his own now," "Akia is really fun and great to have a conversation with!," and "Joseph is really outgoing and funny. I think his conversation skills are good and I genuinely enjoy talking to him."

Postintervention questionnaires were completed by conversational partners who had five or more interactions with a participant to assess their perceptions of the effectiveness of communication book use (see Table 4). Ratings indicated that partners generally believed that participants were initiating and responding to conversation more and that their conversation books helped them talk "some of the time." Finally, normative data collected on 135 general education dyads across 9 classroom

Table 3
Conversational Partner Perceptions

	Did you enjoy this interaction?	Would you like to have this kind of interaction again?	Do you think your partner enjoyed this interaction?	Do you have similar interactions with your friends?	Did you feel that your partner interacted with you appropriately?	Overall *M*
Conner (17)[a]	4.5[b] (4–5)[c]	4.7 (3.5)	4.5 (3.5–5)	4.2 (4–5)	5.0 (4.5–5)	4.58
Joseph (20)	4.2 (2–5)	4.5 (3–5)	3.7 (2–5)	3.6 (2–5)	4.8 (4–5)	4.16
Gabby (14)	4.2 (3–5)	4.4 (3–5)	4.0 (3–5)	3.0 (1–4)	3.8 (3–5)	3.88
Akia (9)	3.1 (2–5)	3.7 (3–5)	2.8 (2–4)	2.4 (1–5)	3.8 (2–5)	3.16
Lucas (9)	4.8 (4–5)	4.7 (3–5)	4.6 (4–5)	4.0 (1–5)	4.9 (4–5)	4.6
Overall *M*	4.22	4.46	3.95	3.53	4.52	

Note. [a]Number of ratings in parentheses. [b]Mean. [c]Range. Ratings were on a scale of 1 (*not all, never*) to 5 (*a lot, yes*).

settings attended by participants revealed an overall mean of 31% (SD = 31) of time spent interacting during observational sessions, which provided an indication of opportunity to interact in high school classroom settings under naturalistic conditions.

Discussion

In this study, communication book use combined with opportunities to interact was associated with increases in conversational initiations and responses of five high school students identified with intellectual disabilities and autism with their general education peers across school settings. Although participants attended general education classes and ate lunch with their peers on a daily basis, prior to introduction of the intervention, interactions with peers only occurred rarely. Furthermore, no teacher or school staff prompted interaction between participants and their general education classmates. Following intervention, all participants indicated that they met their preintervention social goals of having more friends at school. In addition, general education conversational partners generally indicated that they enjoyed interacting with participants and they perceived that participants had increased their conversational skills. Peers' perceptions of their conversations with participants were associated, to some extent, with participants' affect and reciprocity during interactions. Our findings extend the literature on communication book use with students with limited social skills in several important ways and add to the few published social skills interventions involving high school students with intellectual disabilities and autism. Furthermore, our study extends findings to a new population of students with more varied communication characteristics. In addition, normative data suggest the extent to which general education dyads have opportunities for and typically interact in classroom settings.

First, communication books—all of which were composed of the same pool of conversational topics and questions—were found to be effective at increasing social interaction across participants who demonstrated a wide range of functional communication behaviors. Specifically, two participants could read questions in their books to their partners and verbally expand on topics discussed. In contrast, two participants were nonverbal and interacted by pointing to questions in their books and nodding

Table 4
Postintervention Intervention Partner Responses

Participant	___ is initiating more conversation now.	___ responds more now when I talk to him/her.	___ talks to me more now.	___'s picture book helps him/her talk more.	I like talking to ___ more now.	The picture book works well to help ___ talk more.
Conner (3)[a]	4[b]	4.5	5	3	4.5	2.5
	Partner 1 comment: "The books help to a degree. They only really give us a topic to talk about. Our conversations go much better when we have a practical, realistic event to talk about."					
Joseph (3)	4	5	5	3.5	4.5	3.5
	Partner 1 comment: "He did a lot better than early this year, had more of a conversation and talked more than just about the cards."					
	Partner 2 comment: "Some of the picture books have the same question twice or maybe 3 times. Sometimes it's the same question asked with different words. They need more variety in them."					
Gabby (3)	3	5	4	3	5	5
	Partner 1 comment: "Gabby is very nice and enjoyable to talk to."					
Akia (2)	3	4. 5	3	3	3.5	4
	Partner 1 comment: "I really wanted Akia to answer with a sentence instead of answer Yes or No. I know it would be difficult for her to answer in a sentence. I wanted to encourage her to talk more and have more conversation with people or me."					
Lucas (3)	3.5	4	3	4	3.5	4

Note. [a]Number of ratings in parentheses. [b]Mean. Ratings were on a scale of 1 (*strongly disagree*) to 5 (*strongly agree*).

or signing in response to partners' questions, whereas a fifth participant also did not read but interacted verbally with partners. Nevertheless, identical teaching methods were used to instruct all participants to use their communication books, and similar effects were found across participants and their conversational partners. Furthermore, postintervention interviews with partners indicated that they did not find the books to be stigmatizing when interacting with participants. Findings suggest the flexibility and adaptability of the books and instructional methods across a range of students. Extended use of the books may reveal their usefulness across an even wider range of students, such as students with Asperger's syndrome or emotional disorders who have limited social interaction skills.

As suggested by Gaylord-Ross et al. (1984), the communication books may have served as a "social 'prosthetic' to facilitate interaction among peers who ordinarily had no common language or cultural base on which to build interactions" (page 246). Because participants rarely interacted with their peers at school even when in proximity, they typically did not share common experiences on which to develop joint conversational topics. However, because our communication books comprised socially valid topics as derived from diverse groups of high school students when conversing informally at school, topics introduced by participants during communication book use likely were ones to which peers could relate. In fact, several dyad members in our study reported that they did share common interests with each other although they had not previously engaged in or discussed these interests together (e.g., music, sports, and school events).

Second, this study was a systematic replication of Hughes et al. (2000), in which five students with limited communication skills and intellectual disabilities, two of whom also had autism, were taught by general education peers to increase their conversational initiations to a variety of general education partners. Participants with disabilities were also taught to self-prompt their initiations. Although peers had a pivotal role during communication book instruction, their role was diminished during actual conversations with participants following training during the generalization condition. Specifically, Hughes and colleagues instructed generalization partners (none of whom were involved in training) to initiate conversation no more than three times per generalization session, to respond with only one statement to participants' initiations, and not to prompt participants to initiate conversation. Because partners' roles during conversational interaction were limited, reciprocity of interaction was compromised. Partners were instructed to assume a passive role during interaction, simply responding to participants' initiations without expanding on topics or asking reciprocal questions in return. In fact, Hughes et al. (2000) did not report partners' initiations to participants other than as a control measure. In contrast, before partners interacted with participants in the current study, we demonstrated to them how to expand on topics introduced in questions by participants and how to ask participants follow-up questions in return. Therefore, the role of both members of conversational dyads in our study was more reciprocal (see Figures 1–5) and conversations more closely resembled those typical of high school students. Furthermore, partners were encouraged to support participants by acting friendly and prompting participants if they did not turn to a new card in their books.

Third, previous social interaction interventions among secondary students with intellectual disabilities or autism typically have focused either on increasing the initiations of the student with disabilities in dyadic relationships (e.g., Gaylord-Ross et al., 1984; Plienis et al., 1987; Scattone, Tingstrom, & Wilczynski, 2006) or those of the general education peer (e.g., Haring & Breen, 1992; Shukla, Kennedy, & Cushing, 1999; Staub & Hunt, 1993). Our intervention was consistent with the recommendation of Gaylord-Ross et al. (1984) to develop a training package to maximize reciprocity of social exchanges versus simply promoting initiations of either students with disabilities or their general education partners, but not both members of a dyad. Although teaching peers to make overtures to and befriend their classmates with disabilities may foster a more accepting and supportive school climate (e.g., Carter & Kennedy, 2006; Haring & Breen, 1992), failing to teach students with disabilities and limited social skills to initiate interaction with peers may encourage their dependence on others to promote social exchange (Hendricks & Wehman, 2009). Furthermore, students need to learn to initiate speech or gestures that will elicit a response (e.g., Koegel & Koegel, 2006). For example, although students with autism in Scattone et al.'s (2006) study increased their initiations to general education peers, findings revealed that few of their initiations were pivotal in promoting reciprocal social interaction with partners. Specifically, students' initiations often did not result in a response from a peer. In contrast, in our study, reciprocal turn taking typically occurred across conversational partners; as shown on Figures 1–5, participants' initiations and partners' responses tended to correspond systematically within dyads across sessions. That is, although there was variability across participants due to their individual verbal repertoires, when participants' initiations increased during a session, partners' responses typically increased in return and vice versa, suggesting the effectiveness of the intervention package in promoting reciprocal conversational exchanges.

In addition, unsolicited student and teacher reports suggested that our combined intervention package (e.g., expanded partner role, communication book use, and opportunity to interact) resulted in conversational interactions that approximated fairly closely those of typical high school students. For example, following the second day of communication book use with Conner, his conversational partner smiled widely and spontaneously remarked, "This is just like a conversation I have with my

friends!" Several teachers and paraprofessionals independently shared that, following communication book training, they saw Conner initiating for the first time to students in class or the cafeteria and that students were starting to say "Hi" to him in the hall. One paraprofessional observed that in choir class, Joseph had a conversation with a peer "just like you and I are having" and that he had never seen this before. In contrast to baseline sessions in which Lucas was often observed covering his head with his jacket hood or scowling when approached, Lucas's art teacher remarked during communication book use, "I think Lucas is getting the most out of this project—he's happy talking!"

Fourth, the role of conversational partners was further expanded beyond that in Hughes et al.'s (2000) study with respect to participants in our study who were nonverbal or who did not read questions when interacting (i.e., Gabby, Akia, and Lucas). Specifically, during communication book use, we instructed partners to prompt participants to turn pages in their books if they did not do so independently, which partners did intermittently with Gabby and Akia. Perhaps relatedly, partners' initiating to participants was found to be higher with students who needed more support because of limited verbal skills or pointing to rather than reading questions. Partners' affect was also rated more positively, and they were judged to have a greater role in guiding conversations when they interacted with participants requiring more support, perhaps indicating that they enjoyed these conversations more than with students who could read their questions independently and were rated as having a greater role in guiding interactions, or these findings may simply indicate that partners had to "try harder" to get an overt response from students with limited communication repertoires.

Nevertheless, partners' subjective evaluations obtained immediately following conversational interaction sessions were generally more positive with students who were more verbal. Partners' overall ratings were the lowest with Akia, who never spoke and who maintained a flat affect throughout all interactions. It may be that partners, unintentionally or not, were modeling positive social behaviors to engage participants with limited verbal skills and promote their conversational interactions. However, when asked if they enjoyed an interaction or would like to have such an interaction again, partners gave more affirmative ratings to participants who were more verbal. Likewise, partners indicated that interactions with verbal participants were more similar to those with their friends and that these participants interacted more appropriately. For example, the only students with whom some partners responded "Never" when asked, "When you are with your friends, do you have similar interactions?" were those requiring more support when interacting. Although all participants initiated conversational topics within the expected range established by Hughes et al. (1995) for high school students interacting socially, it is not likely that partners had extensive experience interacting with peers who were nonverbal and pointed to rather than verbally asked conversational questions.

These findings suggest that a more extensive disability awareness and orientation should have been provided to partners before interacting with participants. For example, in a postintervention interview, one conversational partner said, "I really wanted Akia to answer with a sentence instead of just 'Yes' or 'No,'" likely without realizing the full extent that autism, an intellectual disability, a hearing impairment, and a limited signing vocabulary had on her communication skills. In contrast, Joseph had a long history of talking in lengthy monologues while appearing disinterested in the responses of conversational partners. Providing more information about participants' individual characteristics and communication skills and alerting peers to the fact that some students had limited response repertoires and likely would never interact like typical high school students may have encouraged partners to be more accepting of individual differences and broadened their view of "appropriateness" of interacting. As suggested by Carter, Sisco, Chung, and Stanton-Chapman (2010), interactions may be enhanced when general and special education students are provided information and skills needed to interact competently with each other. In addition, more extensive interacting between individual partners and participants over time may have increased partners' acceptance of individual differences as dyads became more familiar with each other.

On the other hand, a large number of general education peers willingly and repeatedly volunteered to interact with and provide support to students with communication and social interaction needs and, with little training, did have conversational interactions with extended turn taking. In addition, considering that little disability sensitivity training was provided and that conversational partners had little previous interaction with participants, peer perceptions as self-reported in questionnaires and interviews were generally positive and served as a measure of social validation of the effectiveness of the intervention.

Several limitations of the study warrant further attention. First, during communication book use generalization sessions, we set up conversational encounters between participants and their general education peers as a component of the package intervention. Although students were in proximity in class or the lunchroom and could have conversed with each other during baseline, this rarely happened. Providing the opportunity to interact by (a) previously instructing members of the dyad on their respective roles, (b) obtaining their consent to interact, and (c) handing the communication book to the partner to give to the participant likely signaled the dyad to interact. Throughout all conversational sessions, however, no prompting or feedback was provided to the dyads. Despite lack of assistance during conversations, it could be argued that interactions were contrived and not naturalistic. However, observational studies show that without

social programming, little interaction occurs between students with intellectual disabilities and autism and their general education peers (Carter et al., 2005; Hughes et al., 1999; Mu et al., 2000), as corroborated by our own baseline data. In addition, our general education normative data showed that, among general education dyads, opportunities to interact were fairly minimal; conversational interaction in classroom settings occurred on average less than one third of the time, although variability existed across dyads. Furthermore, because studies show that interactive, cooperative grouping rarely occurs in high school general education classrooms (e.g., Carter et al., 2008; Wallace, Anderson, Bartholomay, & Hupp, 2002), it is necessary to program for social interaction to occur.

Second, no data were reported on whether conversational interactions occurred outside communication book use (generalization) sessions or if conversation occurred without the use of communication books. We did note anecdotally that on one occasion, Conner was seated across from a familiar partner when his book was not available. After an initial greeting from the partner, Conner looked down and appeared to be "at a loss for words," did not initiate, and then said "Cuse me," and moved to another table. Because of the limited verbal repertoires of participants, it is likely that communication books or another type of social prosthetic (Gaylord-Ross et al., 1984) would be needed to sustain participants' interactions over time. Providing books outside generalization sessions may have increased participants' interactions across more naturalistic situations.

Third, although conversational "openers" (questions) comprising communication books were drawn from a pool of over 100 student-derived examples and communication books were varied across participants and sessions, there was similarity of questions across books (e.g., "How's your day going?" or "What are you doing after school today?"). In fact, one of Joseph's conversational partners commented in her postintervention interview that "some books have the same question asked with different words. They need more variety in them." This concern was mentioned only with more verbal participants (Conner and Joseph), perhaps because partners had higher expectations for having a typical conversation with them. Over time, continued opportunities to converse with partners may have resulted in participants learning to expand more on topics and ask related questions, although such a change would be unlikely with nonverbal students like Gabby or Akia.

Fourth, conversational encounters between participants and their general education peers were facilitated by researchers versus teachers or other school staff. Although teachers reported great interest in expanding students' social skills and conversational interactions, they were never observed to implement social programming on their own-a finding similar to observational reports conducted in other high school settings (e.g., Carter et al., 2005, 2008; Hughes et al., 1999). Because teachers report (a) valuing social interaction between students with and without disabilities (e.g., Carter & Hughes, 2006) and (b) believing that programming such interaction is feasible (Carter & Pesko, 2008), future efforts should investigate how teachers can be motivated to and reinforced for implementing social interaction interventions. Furthermore, although participants were taught to initiate conversation when opportunities to interact were provided, they were not taught to approach peers to initiate an interaction—a skill that should be addressed in future social interaction studies.

Conversational skills instruction is a first step toward increasing social interaction among students with intellectual disabilities and autism and their peers (e.g., Alwell & Cobb, 2009). By itself, however, instruction is not enough. Our normative data show that even among general education students, little opportunity for social interaction typically occurs in a high school classroom. Creating opportunities to interact socially with peers and providing peer support training must be part of any intervention to teach social skills to students with intellectual disabilities and autism. If we expect these skills to persist and result in sustained interaction as the norm versus the exception in high school, we must address change on three fronts: students with disabilities, their general education peers, and environmental opportunities.

References

Alwell, M., & Cobb, B. (2009). Social and communicative interventions and transition outcomes for youth with disabilities: A systematic review. *Career Development for Exceptional Individuals*, *32*, 94–107. doi:10.1177/0885728809336657

Carter, E. W., & Hughes, C. (2005). Increasing social interaction among adolescents with intellectual disabilities and their general education peers: Effective interventions. *Research and Practice for Persons with Severe Disabilities*, *30*, 179–193.

Carter, E. W., & Hughes, C. (2006). Including high school students with severe disabilities in general education classes: Perspectives of general and special educators, paraprofessionals, and administrators. *Research and Practice for Persons with Severe Disabilities*, *31*, 174–185.

Carter, E. W., Hughes, C., Guth, C. B., & Copeland, S. R. (2005). Factors influencing social interaction among high school students with intellectual disabilities and their general education peers. *American Journal on Mental Retardation*, *110*, 366–377.

Carter, E. W., & Kennedy, C. H. (2006). Promoting access to the general curriculum using peer support strategies. *Research and Practice for Persons with Severe Disabilities*, *31*, 284–292.

Carter, E. W., & Pesko, M. J. (2008). Social validity of peer interaction intervention strategies in high school classrooms: Effectiveness, feasibility, and actual use. *Exceptionality*, *16*, 156–173. doi:10.1080/09362830802198427

Carter, E. W., Sisco, L. G., Brown, L., Brickham, D., & Al-Khabbaz, Z. A. (2008). Peer interactions and academic engagement of youth with developmental disabilities in inclusive middle and high school classrooms. *American Journal on Mental Retardation*, *113*, 479–494. doi:10.1352/2008.113:479-494

Carter, E. W., Sisco, L. G., Chung, Y., & Stanton-Chapman, T. L. (2010). Peer interactions of students with intellectual disabilities and/or autism: A map of the intervention literature. *Research and Practice for Persons with Severe Disabilities*, *35*, 63–79.

Davis, K. M., Boon, R. T., Cihak, D. F., & Fore, C., III (2010). Power cards to improve conversational skills in adolescents with Asperger syndrome. *Focus on Autism and Other Developmental Disabilities*, *25*, 12–22. doi:10.1177/1088357609354299

Fey, M. E. (1986). *Language intervention with young children*. San Diego, CA: College-Hill Press.

Gaylord-Ross, R. J., Haring, T. G., Breen, C., & Pitts-Conway, V. (1984). The training and generalization of social interaction skills with autistic youth. *Journal of Applied Behavior Analysis*, *17*, 229–247. doi:10.1901/jaba.1984.17-229

Gresham, F. M., Sugai, G., & Horner, R. H. (2001). Interpreting outcomes of social skills training for students with high-incidence disabilities. *Exceptional Children*, *67*, 331–344.

Haring, T. G., & Breen, C. G. (1992). A peer-mediated social network intervention to enhance the social integration of persons with moderate and severe disabilities. *Journal of Applied Behavior Analysis*, *25*, 319–333. doi:10.1901/jaba.1992.25-319

Hendricks, D. R., & Wehman, P. (2009). Transition from school to adulthood for youth with autism spectrum disorder: Review and recommendations. *Focus on Autism and Other Developmental Disabilities*, *24*, 77–88. doi:10.1177/1088357608329827

Horner, R. D., & Baer, D. M. (1978). Multiple-probe technique: A variation of the multiple baseline. *Journal of Applied Behavior Analysis*, *11*, 189–196. doi:10.1901/jaba.1978.11-189

Hughes, C., Harmer, M. L., Killian, D. J., & Niarhos, F. (1995). The effects of multiple exemplar self-instructional training on high school students' generalized conversation interactions. *Journal of Applied Behavior Analysis*, *28*, 201–218. doi:10.1901/jaba.1995.28-201

Hughes, C., Rodi, M. S., Lorden, S. W., Pitkin, S. E., Derer, K. R., Hwang, B., et al. (1999). Social interactions of high school students with mental retardation and their general education peers. *American Journal on Mental Retardation*, *104*, 533–544.

Hughes, C., Rung, L. L., Wehmeyer, M. L., Agran, M., Copeland, S. R., & Hwang, B. (2000). Self-prompted communication book use to increase social interaction among high school students. *The Journal of the Association for Persons With Severe Handicaps*, *25*, 153–166.

Hunt, P., Alwell, M., Goetz, L., & Sailor, W. (1990). Generalized effects of conversational skill training. *The Journal of the Association for Persons With Severe Handicaps*, *15*, 250–260.

Individuals with Disabilities Education Improvement Act of 2004, PL 108-446, 20 U.S.C. §§ 1400 *et seq*.

Kazdin, A. E. (1982). *Single case research design: Methods for clinical and applied settings*. New York: Oxford University Press.

Koegel, R. L., & Koegel, L. K. (2006). *Pivotal response treatments for autism: Communication, social, and academic development*. Baltimore: Paul H. Brookes.

Mayer-Johnson (2008). *Boardmaker® with speaking dynamically Pro® [computer software]*. Solana beach, CA: Mayer-Johnson.

Mu, K., Siegel, E., & Allinder, R. (2000). Peer interactions and sociometric status of high school students with moderate or severe disabilities in general education classrooms. *The Journal of the Association for Persons With Severe Handicaps*, *25*, 142–152.

Naraian, S. (2010). "Why not have fun?": Peers make sense of an inclusive high school program. *Intellectual and Developmental Disabilities*, *48*, 14–30. doi:10.1352/1934-9556-48.1.14

No Child Left Behind Act of 2001, PL 107-110, 115 Stat. 1425, 20 U.S.C. §§ 6301 *et seq*.

Plienis, A. J., Hansen, D. J., Ford, F., Smith, S., Jr., Stark, L. J., & Kelly, J. A. (1987). Behavioral small group training to improve the social skills of emotionally-disordered adolescents. *Behavior Therapy*, *18*, 17–32. doi:0005-7894/87/0017-0032

Reichow, B., & Volkmar, F. R. (2010). Social skills interventions for individuals with autism: Evaluation for evidence-based practices within a best evidence synthesis framework. *Journal of Autism and Developmental Disorders*, *40*, 149–166. doi:10.1007/s10803-009-0842-0

Scattone, D., Tingstrom, D. H., & Wilczynski, S. M. (2006). Increasing appropriate social interactions of children with autism spectrum disorders using Social Stories™. *Focus on Autism and Other Developmental Disabilities*, *21*, 211–220. doi:10.1177/10883576060210040201

Schnorr, R. F. (1990). "Peter? He comes and goes…": First-graders' perspectives on a part-time mainstream student. *The Journal of the Association for the Severely Handicapped*, *15*, 231–240.

Shukla, S., Kennedy, C. H., & Cushing, L. S. (1999). Intermediate school students with severe disabilities: Supporting their social participation in general education classrooms. *Journal of Positive Behavior Interventions*, *1*, 130–140. doi:10.1177/109830079900100301

Staub, D., & Hunt, P. (1993). The effects of social interaction training on high school peer tutors of schoolmates with disabilities. *Exceptional Children*, *60*, 41–57.

Test, D. W., Mazzotti, V. L., Mustian, A. L., Fowler, C. H., Kortering, L., & Kohler, P. (2009). Evidence-based secondary transition predictors for improving postschool outcomes for students with disabilities. *Career Development for Exceptional Individuals*, *32*, 160–181. doi:10.1177/0885728809346960

U.S. Department of Education. (2009). *28th Annual report to Congress on the Implementation of the Individuals With Disabilities Education Act, 2006*. Washington, DC: Author.

Wallace, T., Anderson, A. R., Bartholomay, T., & Hupp, S. (2002). An ecobehavioral examination of high school classrooms that include students with disabilities. *Exceptional Children*, *68*, 345–359.

Wang, P., & Spillane, A. (2009). Evidence-based social skills interventions for children with autism: A meta-analysis. *Education and Training in Developmental Disabilities*, *44*, 318–342.

Wolf, M. M. (1978). Social validity: The case for subjective measurement or how applied behavior analysis is finding its heart. *Journal of Applied Behavior Analysis*, *11*, 203–214.

Received: December 23, 2010
Final Acceptance: March 23, 2011
Editor in Charge: Fred Spooner